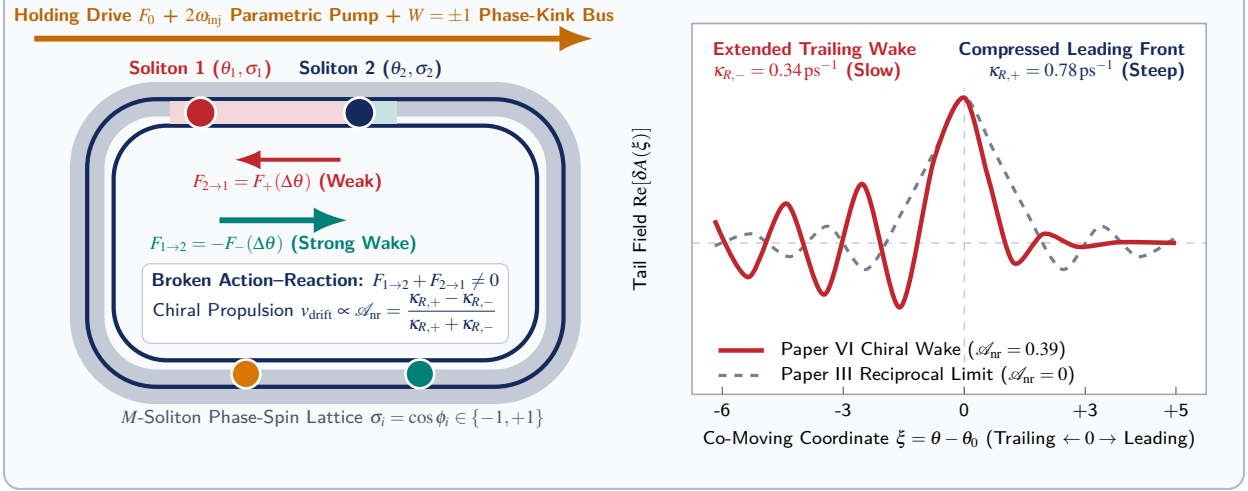
(a) Driven Ring Resonator with Convective Wake ($\kappa_+ \neq \kappa_-$)
Holding Drive F_0 + $2\omega_{\text{inj}}$ Parametric Pump + $W = \pm 1$ Phase-Kink Bus
Soliton 1 (θ_1, σ_1)
Soliton 2 (θ_2, σ_2)
$F_{2\to1} = F_+(\Delta\theta)$ (Weak)
$F_{1\to2} = -F_-(\Delta\theta)$ (Strong Wake)
Broken Action–Reaction: $F_{1\to2} + F_{2\to1} \neq 0$
Chiral Propulsion $v_{\text{drift}} \propto \mathcal{A}_{\text{nr}} = \frac{\kappa_{R,+} - \kappa_{R,-}}{\kappa_{R,+} + \kappa_{R,-}}$
M-Soliton Phase-Spin Lattice $\sigma_i = \cos\phi_i \in \{-1, +1\}$
(b) Asymmetric Soliton Tail Profile vs. Paper III
Extended Trailing Wake
$\kappa_{R,-} = 0.34\,\text{ps}^{-1}$ (Slow)
Compressed Leading Front
$\kappa_{R,+} = 0.78\,\text{ps}^{-1}$ (Steep)
Tail Field Re[$\delta A(\xi)$]
Paper VI Chiral Wake ($\mathcal{A}_{\text{nr}} = 0.39$)
Paper III Reciprocal Limit ($\mathcal{A}_{\text{nr}} = 0$)
-6
-3
0
+3
+5
Co-Moving Coordinate $\xi = \theta - \theta_0$ (Trailing $\leftarrow 0 \rightarrow$ Leading)